

ArrayStack.java. (generic stack array)

data = [null, null, null, null]
top = -1

push = 10
[10, null, null, null]
↑
Top = 0 (+ + Top)

push = 20
[10, 20, null, null]
↑
Top = 1

push = 30
[10, 20, 30, null]
↑
Top = 2

Peek → 30 peek()

pop → 30
[10, 20, ~~30~~, null, null]
↑
Top = 1

push → 40
[10, 20, 40, null]
↑
Top = 2

push → 50 [10, 20, 40, 50]
↑
Top = 3

push = 60

is full (top = arr.length - 1) ?

→ stack overflow

pop → 50

pop → 40

pop → 30

pop → 20

pop → 10

pop → is Empty top = -1

→ Stack underflow

arr = [50, 40, 30, 20, 10] → arr = [50, 40, 30, 20]
 arr = [50, 40, 30, 20]

↓ move to 21 front - 8

[0, 0, 0, 0, 0, 0, 0, 0] = 0
 (index) 0 = 0

[0, 0, 0, 0, 0, 0, 0, 0] → [5] → [5]
 [0, 0, 0, 0, 0, 0, 0, 0] → [5] → [5]
 [0, 0, 0, 0, 0, 0, 0, 0] → [5] → [5]

21 - 670
 [0, 0, 0, 0, 0, 0, 0, 0]
 2 = 2212
 → 1112